Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification

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Assigned System: Warehouse Inventory & Pallet Location Tracker

Architecture Selection: I chose Layered Architecture for the Warehouse Inventory & Pallet Location Tracker System.

Reason 1: Clear Separation of Warehouse Operations

The system has three main areas that fit well into different layers. The Presentation Layer handles scans from handheld or touch-screen devices used in the warehouse. The Business Layer handles things like stock updates, changing pallet locations and generating reports. The Data Layer stores inventory records, pallet locations and pricing data. This separation keeps the user interface simple and keeps the database details separate from the main application logic. It also makes the system easier for a small team to understand, test and maintain.

Reason 2: Predictable, Centralised Data Consistency

A warehouse should not have two different records for the location of the same pallet. In the layered design, the Business Layer handles the requests before they reach the Data Layer, which uses one Warehouse Database Component. This keeps inventory counts and pallet locations consistent. It also avoids some of the data synchronisation issues that could come with a distributed design such as Microservices. For a system of this size, a layered design is also simpler to deploy and manage.

Security Advantage

The layered structure also helps with security. The Scanner/Warehouse UI does not directly access the Warehouse Database. Requests go through the Business Layer, which includes the Inventory Manager, Pallet Location Tracker and Report Generator. This gives the Business Layer a place to check scanned input, permissions and user roles before data reaches the database. It can also help stop incorrect or harmful requests from directly reaching the database.

Performance Benefit

Separating the layers also makes it easier to improve performance. For example, the Business Layer can temporarily keep frequently used inventory and pallet-location data in memory. This can reduce repeated database queries when the same pallet or SKU is scanned many times. The Data Layer can also use indexes for pallet ID and location searches. These changes can be made without having to change the user interface.

Identified Components

1. Scanner / Warehouse UI Component — Presentation Layer

2. Inventory Manager Component — Business Layer

3. Pallet Location Tracker Component — Business Layer

4. Report Generator Component — Business Layer

5. Warehouse Database Component — Data Layer

Component Diagram

